

- 3 A student carried out an experiment to determine the resistivity ρ of copper using a copper wire. The uncertainties in the measurements are shown.

uncertainty in length l of wire = 0.2%

uncertainty in diameter d of wire = 1.6%

The equation for resistivity ρ is $\rho = \frac{\pi d^2 R}{4l}$.

He obtains a resistivity value of $(1.71 \pm 0.07) \times 10^{-8} \Omega \text{ m}$ with its associated uncertainty.

What is the uncertainty in the measurement of resistance R of the wire?

A 0.007%

B 0.7%

C 0.9%

D 7%

Ans: B

$$\frac{0.07}{1.71} = 2 \left(\frac{1.6}{100} \right) + \frac{\Delta R}{R} + \frac{0.2}{100}$$

$$\frac{\Delta R}{R} = 0.007 = 0.7\%$$

5

Option A: If student forgets to convert to percentage.

Option C: If student includes 4 in percentage uncertainty to give $0.0009 = 0.9\%$

Option D: If student makes R the subject first and adds all the percentage uncertainty.